

Exhibit*Guide*

A smart audio & content guide for museums and galleries. Visitors scan the exhibit label with their own phone — no hardware rental, no app install, no account — and get the story behind the work, their way.

Progressive Web App

QR scan — no app install

B2B2C — institutions subscribe

I. SCAN

A QR code on any exhibit label opens the page instantly — no app, no install.

II. READ OR LISTEN

A 30-second TLDR up front; text, audio, video and imagery on demand.

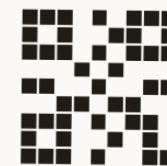
III. CATALOGUE

Optionally save exhibits to a personal collection to revisit after the visit.

JOHANNES VERMEER

*Girl with a Pearl
Earring, c. 1665*

Oil on canvas · 44.5 × 39 cm



Scan for the story —

*thirty seconds, or as
deep as you like.*

Visitors arrive curious and connected – the *storytelling* hasn't caught up.

Galleries are busier than ever and nearly every visitor already carries a capable smartphone. Yet the tools to explain the work are still a printed label of forty words or a rented handset behind a queue. **The phone in the pocket is the opportunity.**

~90%

of gallery visitors carry a smartphone capable of scanning a QR code.

Illustrative — smartphone penetration

1 in 5

visitors rent an audio handset; cost, queues and hygiene deter the rest.

Illustrative — handset uptake

~60%

say wall labels are too short – they want more depth, on their terms.

Illustrative — visitor surveys

2_{sec}

is all the patience a visitor gives a page mid-visit before moving on.

Web performance benchmarks

THREE FORCES CONVERGING

QR went mainstream

Menus and tickets normalised scanning – no instruction needed anymore.

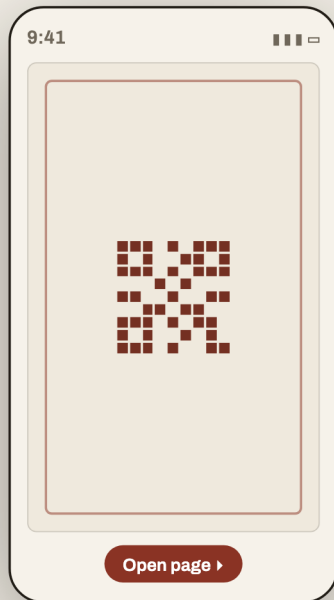
App fatigue is real

Nobody installs a single-visit app or signs up at the gallery door.

Institutions want reach

Curators have rich content but no engineering team to publish it.

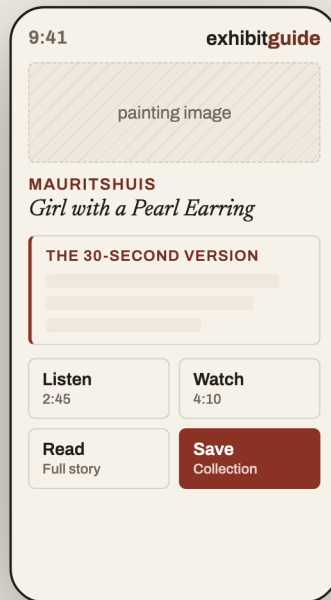
From the painting to a saved piece in *four taps*.



i.

Scan

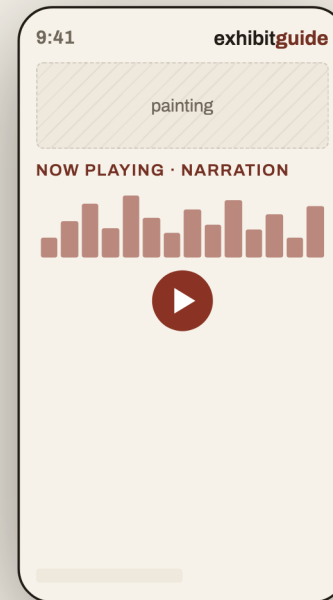
Camera reads the QR on the mount — no app, no install.



ii.

Choose

TLDR up top; listen, watch, read or save — all without signing in.



iii.

Listen

Hands-free narration so eyes stay on the work, not the phone.



iv.

Save

The only sign-up gate — and only once the value is clear.

What it actually *feels like* to use, start to finish.



Maya, 28, is on a weekend city break with an hour spare at the Mauritshuis. She loves art but isn't an expert, her phone is nearly full, and she'd never download an app for a single visit.

14:02 · STANDS

In front of the work

The printed label gives forty words. She wants the story, not a queue for a handset.

14:03 · SCANS

Her own phone

Camera reads the QR on the mount. One tap opens the page — no app, no sign-up.

14:03 · SKIMS

The 30-second TLDR

She gets the gist instantly, then taps to go deeper on the parts that intrigue her.

14:06 · LISTENS

Hands-free narration

Audio plays in her earphones so her eyes stay on the painting, not the screen.

19:40 · REVISITS

Saved, on the train home

She'd tapped "save" — now reads the full provenance and keeps it in her collection.

THE USP

No rental counter, no app store, no sign-up wall — and yet more depth and a record to take home than any current option offers.

Three kinds of visitor shaped every *design choice*.

Maya 28

THE TIME-PRESSED TOURIST

DEMOGRAPHICS

International visitor, short stay, mobile-first, on patchy gallery WiFi.

PSYCHOGRAPHICS

Curious but time-poor; wants the gist, not a guided tour or commitment.

BEHAVIOUR

Scans, skims the TLDR, listens while looking, moves on within minutes.

David 54

THE LIFELONG LEARNER

DEMOGRAPHICS

Local member, frequent visitor, comfortable with technology and depth.

PSYCHOGRAPHICS

Enjoys context and technique; values quality content over novelty.

BEHAVIOUR

Lingers, plays full audio, watches the film, returns across many visits.

Aisha 23

THE STUDENT & RESEARCHER

DEMOGRAPHICS

Art-history student; visits with academic intent and a reading list.

PSYCHOGRAPHICS

Rigorous; needs sources she can trust, quote and revisit later.

BEHAVIOUR

Saves pieces, attaches notes, exports citation-ready records after the visit.

The curator

SECONDARY — THE PAYING CUSTOMER

Small-to-mid institutions with rich content but no engineering team. They publish exhibits from a simple CMS and subscribe per institution — the people who shape demographics, psychographics *and* the business model alike.

A boring, proven stack — chosen so we ship the *product*, not the plumbing.

Frontend

React PWA + Tailwind

Mobile-first and installable-but-not-required. A service worker caches pages so exhibits load on patchy gallery WiFi — and there is no app store in the way.

API

Django REST Framework

One clean, versioned REST layer serving both the visitor PWA and the operator CMS — well-documented, fast to build, easy to test.

Backend + CMS

Django + admin

Batteries-included Python. The Django admin *is* the MVP CMS, so curators publish on day one with zero custom build (Wagtail comes later).

Auth

dj-rest-auth + JWT

Register, login, logout and password reset out of the box — token auth that suits a stateless PWA. SSO is deferred until needed.

Database

PostgreSQL

A single reliable store for exhibits, accounts, collections, annotations and anonymised scan events — relational, queryable, boring in the best way.

Media

S3-compatible + CDN

Audio, video and images off the app server and onto a CDN. Audio kept under ~1 MB so narration starts fast inside the gallery.

DEFERRED BY DESIGN

Held back until the core is validated — added only when a real need is proven: **Celery** + **Redis** async jobs · **Claude API** for AI summaries & translation · **Wagtail** richer CMS · **SSO** (Google / Apple).

Agile in one-week sprints — version-controlled and *demoable every week.*

Development method **AGILE — SCRUM FLAVOUR**

Wireframes mapped to user stories; a prioritised feature backlog; a demoable increment presented every Thursday, with UAT in a real gallery from Sprint 2.

Version control **GITHUB**

Feature-branch workflow — each branch maps to a backlog item and sprint deliverable; pull-request review; main always deployable.

THE WEEKLY LOOP

i.

Plan

Pull the next user stories off the prioritised backlog into the sprint.

ii.

Build

One feature branch per story; PR review; tests run in CI before merge.

iii.

Demo

A working increment shown every Thursday for feedback and re-prioritising.

iv.

Validate

From Sprint 2, test the increment with real visitors in a real gallery.

DEFINITION OF DONE

Merged via PR review · automated tests green in CI · deployed to staging · and demoed against its user story's acceptance criteria.

One MVP, two halves: read free, then *save · annotate · manage*.

READ CORE

MVP

Scan to story — no login

The validated core — built and piloted first

- **QR scanning** — phone camera opens the PWA page directly
- **Exhibit page** — 30-second TLDR + expandable full content
- **READ in 3 formats** — text, audio narration and short film
- **Format preference** — remembered across exhibits
- **PWA + offline** — cached for patchy gallery WiFi
- **Operator CMS** — curators publish via the Django admin

ACCOUNTS & COLLECTIONS

MVP · CRUD

Save, annotate, manage

The account gate appears only at the moment of saving

- **Accounts** — email register, login, logout and password reset
- **Create** — save an exhibit into a personal collection
- **Read** — re-open saved content after the visit
- **Update** — annotate saved pieces; rename collections
- **Delete** — remove a saved exhibit or a whole collection

OUT OF MVP SCOPE

Deliberately held back until the core proves itself: **AI layer** (summaries, translation, related works) · **accessibility mode** · **operator analytics** · **SSO** · **export & visit history**.

The full backlog — *MVP epics* first, later phases sequenced behind.

EPIC 1 · 2 MVP Content & READ ■ Operator CMS — publish exhibits ■ QR deep-link to exhibit page ■ 30-second TLDR + full content ■ Text · audio · video formats ■ Format preference, remembered	EPIC 3 MVP Accounts ■ Email register ■ Login / logout ■ Password reset ■ JWT session for the PWA ■ Gate shown only at save	EPIC 4 · 5 · 6 MVP · CRUD Collections ■ Create — save an exhibit ■ Read — revisit post-visit ■ Update — annotate · rename ■ Delete — remove item / set ■ Epic 7 — PWA offline polish	POST-MVP LATER Later phases ■ AI summaries & translation ■ Related-works suggestions ■ Accessibility mode ■ Operator analytics · SSO ■ Export & visit history
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Six people, one gallery, *one scan each*.

QR DEEP-LINK · NO ACCOUNT

As a first-time visitor,

I WANT exhibit pages to open straight from my camera,

SO THAT I never install an app or sign up mid-visit.

TLDR SUMMARY

As a time-pressed tourist,

I WANT a thirty-second summary the moment I scan,

SO THAT I get the story without breaking my stride.

AUDIO · FORMAT PREFERENCE

As a visitor absorbed in the work,

I WANT narration to play in my earphones,

SO THAT I learn without taking my eyes off the piece.

COLLECTIONS · CREATE + UPDATE

As an academic researcher,

I WANT to save exhibits and annotate them with my own notes,

SO THAT I can study what I saw after the visit.

LOGIN · POST-VISIT RETURN

As a returning visitor,

I WANT to log in and re-open my saved collection,

SO THAT I pick up exactly where I left off after the visit.

MUSEUM CMS

As a curator,

I WANT to publish exhibit content from a simple CMS,

SO THAT I update the gallery without engineering help.

Content first, public READ *piloted early*, then auth & collection CRUD.

TUE Kick-off

Project understanding & scope sign-off with the client.

- Brief confirmed
- Backlog seeded

SPRINT 01 Epic 0

Scaffold the foundations the rest of the build sits on.

- Django + DRF + Postgres
- React PWA shell
- Data model · wireframes

SPRINT 02 Epics 1–2

The validated core — and the first real-world test.

- Operator CMS
- Scan → content, 3 formats
- Pilot in a real gallery

SPRINT 03 Epics 3–5

Identity, then the create / read / update of collections.

- Accounts (JWT)
- Save & revisit exhibits
- Annotate · rename

SPRINT 04 Epics 6–7

Complete CRUD, harden the PWA and act on UAT.

- Delete items / sets
- Offline polish
- UAT findings & polish

Two test loops: does the *code* work, and does it work for *visitors*?

TESTING THE CODE

AUTOMATED

- **Unit tests** – pytest for Django / DRF logic; Jest + React Testing Library for components.
- **API & integration** – DRF APIClient covers auth and every collection CRUD endpoint.
- **End-to-end** – Playwright walks the critical path: scan → content → save → revisit.
- **CI gate** – GitHub Actions runs the suite on every pull request; red build blocks merge.
- **Performance** – Lighthouse budgets & Web Vitals guard the < 2-second scan-to-content bar.
- **Quality** – lint, type-checks and a coverage threshold on the core paths.

TESTING WITH USERS

IN THE GALLERY

- **Usability sessions** – think-aloud runs with each persona: tourist, learner, researcher.
- **In-gallery UAT** – from Sprint 2, on real phones, real WiFi and real exhibit labels.
- **Task success** – can a first-timer scan and reach content unaided? Time-on-task tracked.
- **Format A/B** – which of text · audio · video actually drives engagement.
- **Accessibility** – screen-reader pass, contrast and simplified-language checks.
- **Curator test** – can a non-technical curator publish an exhibit with no help?

ACCEPTANCE *A story isn’t “done” until its automated test passes *and* a real user completes the task it describes.*

Engagement proves the visit; *institution renewal* proves the business.

NORTH STAR *Engaged scans per exhibit, per institution — the one number that shows a visitor got value and gives a curator a reason to renew.*

VISITOR VALUE — DOES THE EXPERIENCE WORK?

REACH Scan-through <i>rate</i> % of scans that open the page Source — scan events ÷ page loads	ACTIVATION Scan → content Primary: < 2 s, live gallery Source — Sentry / Web Vitals	DEPTH Format <i>engagement</i> Split: audio · video · text Source — PostHog	LOYALTY Save & <i>return</i> Save conv · post-visit re-open A depth signal — not a vanity count
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BUSINESS VALUE — WHY A MUSEUM INVESTS, AND RENEWS

OPERATOR Publish <i>time</i> Minutes to publish an exhibit Source — CMS timestamps	PROVEN ROI Engagement <i>per exhibit</i> The renewal argument Source — operator dashboard	RETENTION Institution <i>renewal</i> The growth metric (NRR) Source — subscriptions	EXPANSION New sites & <i>referrals</i> Curators talk to curators Source — pipeline / CRM
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THE FLYWHEEL Most visitors come **once** — so chasing consumer DAU or 7-day retention would mislead. Engaged visitors make curators look good; **curators see the ROI and renew**; renewals and referrals grow the network — which brings more visitors.

Instrumented privacy-first: PostHog (self-hosted per tenant) · Plausible (cookieless) · Sentry / Web Vitals · Metabase dashboards for curators.

Thank you — *your turn.*

I'd value feedback on scope and on the friction-free, account-light approach — and approval to begin the Sprint 1 build of the scan-to-story MVP.

FOR YOUR APPROVAL

- i.* Is the MVP scope right?
READ-first core, then full collection CRUD; AI & export deferred.
- ii.* Is the audience well-targeted?
Tourist, learner, researcher — plus the curator who pays.
- iii.* Any concerns on the stack?
React PWA + DRF/Django, QR-first, JWT auth, Django-admin CMS.